

Range Control System Requirements Summary

1. Primary Objective

Build a **web-based range-control application** that allows authorized firearms instructors to control the existing target system, create and run drills, pause/resume drills, and record training results.

The system should automate the existing manual target-control process while remaining flexible enough to support the department's highly customized training exercises.

The department conducts substantially more customized training than standardized qualification drills, including changes to round counts, distances, weapons, movement, and time limits.

2. Drill Creation and Management

The application needs a **drill builder**.

An instructor should be able to:

- Create a new drill.
- Name the drill.
- Save it for later use.
- Edit an existing drill.
- Create a new drill based on an existing one.
- Define multiple stages/commands within a drill.
- Specify the target(s) involved in each stage.
- Specify how long a target remains activated.
- Specify delays/timing between actions.
- Specify the number of rounds/actions.
- Organize several stages into one complete course of fire.

The instructor specifically wants drills saved for reuse rather than recreated each time.

3. Pause / Resume

Pause/resume is a **core requirement**, not merely a convenience.

The instructor needs to pause after a stage so shooters can move to another yard line, top off magazines, check weapon condition, correct a malfunction, or otherwise prepare for the next stage.

The drill must resume from the point where it was paused rather than restarting the entire sequence.

4. Target Control

There are approximately **20 target units**.

Targets should be controllable individually because the existing system uses individual pistons/solenoids.

The application should support selecting individual targets or groups of targets for each drill stage.

The system should account for targets that are broken or unavailable. Instructors discussed marking malfunctioning targets so that an 'all targets' or randomized drill automatically excludes them.

Target activation/deactivation timing must be programmable.

5. Timing

Timing must be configurable for each drill/stage.

Requirements discussed include:

- Start delays.
- Target presentation duration.
- Delays between stages.
- Different time limits for different exercises.
- Manual pause/resume independent of the timer.

Examples discussed include exercises such as six rounds in four seconds, five rounds in ten seconds, and other customized combinations.

6. Preset and Custom Drills

The application should support **both preset and custom drills**.

Preset drills can represent established qualification or commonly used exercises.

Custom drills are particularly important because the department estimates that the majority of its training is customized rather than limited to its formal qualification drills.

7. Turning Targets / Shoot-No-Shoot

If the existing turning-target system can be restored, the application should eventually support different target orientations.

A future training mode discussed is **shoot/no-shoot**, where targets turn in different configurations and the shooter must identify the appropriate response.

This is a desirable enhancement and was discussed as a future capability rather than a clearly defined minimum initial requirement.

8. Users and Access

Approximately **six people** are currently expected to run drills regularly.

Approximately **10 accounts** would likely be sufficient if individual logins are implemented.

The department indicated that controlling access through the existing internal Wi-Fi/password process may be preferable to a complicated account-management system.

9. Multi-Shooter Operations

Training may involve approximately 6–8 shooters at a time, with SWAT sessions reaching approximately 12 shooters.

Multiple firearms instructors may be present, but only **one person gives commands to the firing line** at a time.

Other instructors function as range safety officers.

The application therefore needs to support one instructor controlling the drill while multiple shooters participate.

10. Scoring and Training Records

The department currently records qualification results on paper and manually enters them into an outdated records-management system.

The proposed system should provide a centralized way to:

- Record scores.
- Associate results with individual officers.
- Retrieve an officer's previous results.
- Review historical performance.
- Compare results between training periods.

Approximately **80 officers** was discussed as a reasonable population for stored records.

11. Leaderboard / Performance Tracking

A leaderboard or performance-tracking feature was discussed, particularly for SWAT.

Potential information could include times, scores, and individual historical performance.

The exact scoring methodology for a leaderboard has **not been finalized**.

A shot timer could potentially provide an automated timing component.

12. Shot Timer Integration

The department currently uses a **Shooters Global shot timer**.

The safety officer monitors the shooter and the timer during qualification/training.

A digital shot-timer capability inside the application was discussed as a possible enhancement.

The transcript also describes the existing timer as problematic for this workflow because it does not provide the required pause points between stages.

13. Existing Hardware / Infrastructure

The existing target system is described as old and manufactured by **Action Target**.

Approximately 20 targets exist, with some currently malfunctioning.

Some locking bolts have snapped, preventing certain targets from reliably rotating.

The target tower and surrounding infrastructure have suffered significant deterioration, including damage from a fallen tree.

The existing controller/microcontroller has a limited number of connections.

Some existing components/part numbers may no longer be manufactured.

Electrical damage, including possible lightning-related damage, is a concern.

14. Network / Connectivity

The application is expected to communicate with the target-control hardware over the range's Wi-Fi network.

If Wi-Fi coverage does not reliably reach the target tower, a **Wi-Fi repeater** was identified as a straightforward potential solution.

The system should account for connectivity limitations because the target hardware is physically separated from the range house.

15. Physical Testing and Development Access

Physical access to the range is important for development and testing because the project directly controls physical hardware.

Range personnel indicated that the development team can coordinate visits and obtain access to locked areas as needed.

Pictures, hardware diagrams, relay information, and other documentation can be provided by range personnel. Testing access will need to be coordinated around firearms training and range availability.

16. Out-of-Scope / Future Ideas

A moving-target/track system and a dedicated steel-target/plate-rack area were discussed, but these were explicitly characterized as likely **out of scope** for the current project.

The primary project should focus on controlling the existing target infrastructure and delivering the drill-management functionality.

17. Key Minimum Viable Product

Web-based range control connected to the existing target hardware.

Preset and custom drill creation.

Saved/editable drills.

Individual target selection.

Configurable timing.

Pause/resume between stages.

Broken/unavailable target management.

Basic instructor access control.

Basic score/result recording tied to individual officers.

Reliable physical testing with the existing hardware.

18. Important Open Technical Questions

Which existing relays/controllers can be reused and which need replacement?

Exactly how are the individual target solenoids wired and addressed?

How many targets can the current microcontroller/control board independently actuate? What hardware is currently operational versus damaged?

How should Wi-Fi loss or controller failure be handled safely?

What exact data fields should be stored for a drill and for an officer's results?

What scoring formula, if any, should be used for a leaderboard?

Which turning-target functionality can realistically be restored with the existing mechanical system?